

INDUSTRIAL PRODUCT DATASHEET

# reServer Industrial J4012- Fanless AI-enabled NVR Server with NVIDIA Jetson Orin™ NX 16GB module

Fanless AI-enabled NVR server with NVIDIA Jetson Orin™ NX 16GB module for video analytics and smart storage, with a price increase anticipated in the coming weeks.

SKU

114110247

UPDATED

2026-09-07 07:49:30

PRICE

\$1899.00(Excl. VAT)



# Overview



The reServer Industrial J4012 is a fanless AI-enabled NVR server powered by the NVIDIA Jetson Orin™ NX 16GB module, delivering up to 100 TOPS for real-time multi-stream video processing and smart storage with two 2.5-inch drive bays. Due to surging demand for modules and SSDs, prices for reComputer and reServer Jetson devices are expected to increase by approximately \$50 to \$400 in the coming weeks; please confirm requirements and place orders promptly to secure stock.

# Key Features

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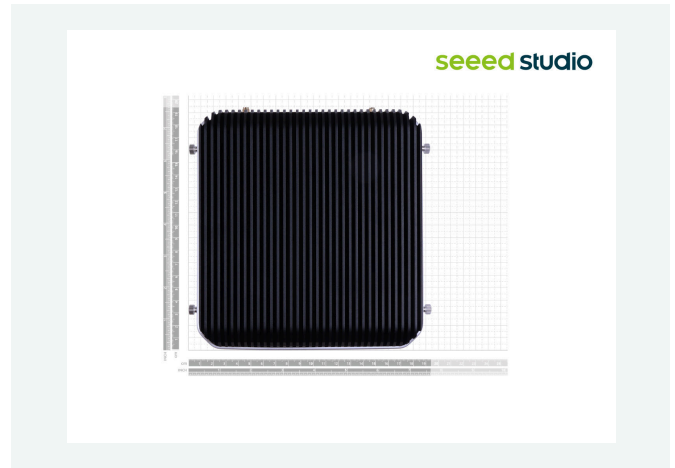
## Price Increase Announcement

**Date: February 28, 2026**

Due to surging demand for modules and SSDs, prices for reComputer and reServer Jetson devices will increase by approximately \$50 to \$400 in the coming weeks. Please confirm your requirements and place your orders promptly to secure your stock.

# Product Gallery



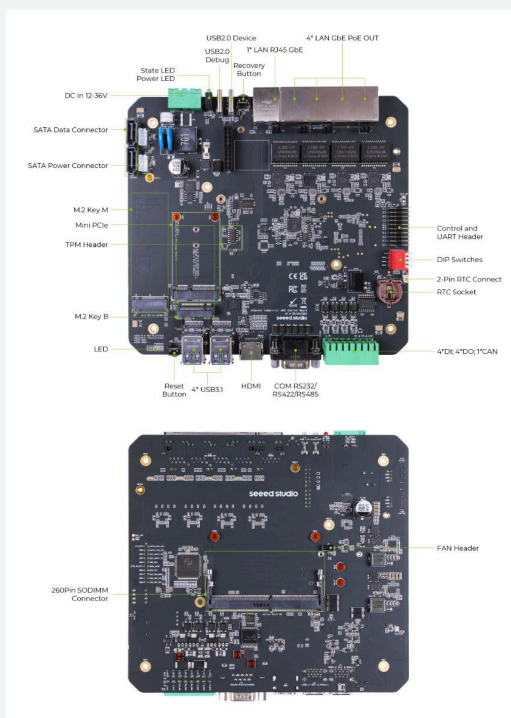
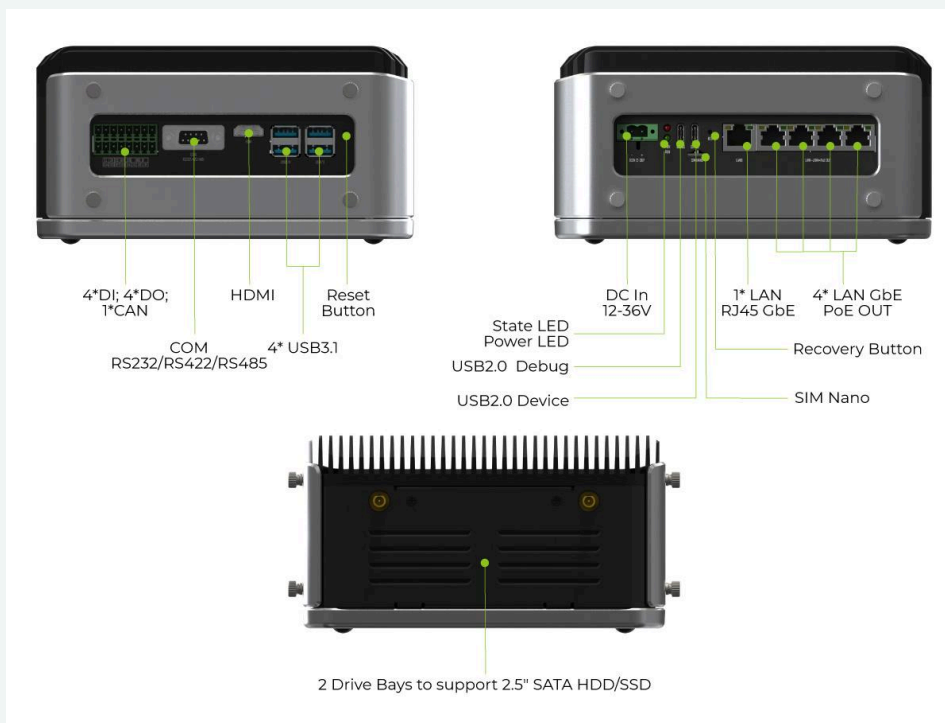


## Specification

| Jetson Orin NX 16GB System on Module |                                                                                                |                                                                                   |
|--------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| AI Performance                       | reServer Industrial J4012 - Orin NX 16GB, 100 TOPS                                             | reServer Industrial J4011- Orin NX 8GB – 70 TOPS                                  |
| GPU                                  | 1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores                                  |                                                                                   |
| CPU                                  | 8-core Arm® Cortex®-A78AE v8.2 64-bit CPU (Orin NX 16GB)                                       | 6-core Arm® Cortex®-A78AE (Orin NX 8GB)                                           |
| Memory                               | 16GB 128-bit LPDDR5 (Orin NX 16GB)                                                             | 8GB 128-bit LPDDR5 (Orin NX 8GB)                                                  |
| Video Encode                         | 1x 4K60 (H.265)   3x 4K30 (H.265)   6x 1080p60 (H.265)   12x 1080p30 (H.265)                   |                                                                                   |
| Video Decode                         | 1x 8K30 (H.265)   2x 4K60 (H.265)   4x 4K30 (H.265)   9x 1080p60 (H.265)   18x 1080p30 (H.265) |                                                                                   |
| DL Accelerator                       | 2x NVDLA v2                                                                                    | 1x NVDLA v2                                                                       |
| Carrier Board                        |                                                                                                |                                                                                   |
| Storage                              |                                                                                                | 1x M.2 Key M PCIe (M.2 NVMe 2280 SSD 128G included)                               |
| I/O                                  | Ethernet                                                                                       | 1x RJ45 GbE (10/100/1000M)<br>4x RJ45 GbE PoE(PSE 802.3 af 15 W, 10/100/1000Mbps) |
|                                      | USB                                                                                            | 4x USB3.1 Type A, 2x USB2.0 Type C                                                |
|                                      | DI/DO                                                                                          | 4x DI, 4x DO, 3x GND_DI, 2x GND_DO, 1x GND_ISO, 1x CAN                            |
|                                      | Display                                                                                        | 1x HDMI 1.4                                                                       |
|                                      | Fan                                                                                            | 1x 4 pin Fan Connector (5V PWM)                                                   |
|                                      | COM                                                                                            | 1x DB9 (RS232/RS422/RS485)                                                        |
|                                      | SATA                                                                                           | 2x Drive Bays to support 2.5" SATA HDD/SSD(SATA III 6.0Gbps)                      |
|                                      | SIM                                                                                            | 1x Nano SIM card slot                                                             |
|                                      | Button                                                                                         | Reset Button, Recovery Button                                                     |
| Expansion                            | RTC                                                                                            | 1x RTC 2-pin, supports CR1220 but not included                                    |
|                                      | Mini PCIe                                                                                      | 1x Mini PCIe for LoRaWAN®/4G/5G/Series Wireless (Module optional)                 |
|                                      | M.2 Key B                                                                                      | 1x M.2 Key B (3042/3052) support 4G/5G (Module optional)                          |
|                                      | Fan                                                                                            | Fanless, passive heatsink; 1x Fan connectors(5V PWM)                              |
|                                      | TPM                                                                                            | 1x TPM 2.0 connector (Module optional)                                            |
| Power                                | DC 12V-36V Terminal block 2 pin                                                                |                                                                                   |

|                       |                        |                                                                 |
|-----------------------|------------------------|-----------------------------------------------------------------|
| Mechanical            | Dimensions (W x D x H) | 194.33mm x 187mm x 95.5mm                                       |
|                       | Installation           | Desk, DIN rail, VESA                                            |
|                       | Weight                 | 2.8kg                                                           |
| Operating Temperature |                        | -20 ~ 60°C with 0.7m/s                                          |
| Operating Humidity    |                        | 95% @ 40 °C (Non-condensing)                                    |
| Storage temperature   |                        | -40 ~ 85°C                                                      |
| Storage humidity      |                        | 60°C@ 95% RH(Non-condensing)                                    |
| Vibration             |                        | 3 Grms @ 5 ~ 500 Hz, random, 1 hr/axis                          |
| Shock                 |                        | 50G peak acceleration (11 msec. duration,eMMC,microSD,or mSATA) |
| Warranty              |                        | 2 Years                                                         |

# Hardware Overview



# Applications

- **Unlock AI capabilities for existing cameras and VMS:** Powered by the NVIDIA Jetson Orin NX 16GB processor, delivering up to 100 TOPS of AI performance, and capable of processing multiple video streams in real-time. The reServer empowers you to apply high-performance filters for critical insights across vast video footage.
- **Smart Storage Solutions:** Store and upload only noteworthy recordings for further processing. With two 2.5-inch hard drive bays, the reServer Industrial provides ample storage capacity, ensuring you have the space needed for expanding data requirements.
- **Create custom analytics made it easy**
- **Out-of-box tailor AI solution:** CVEDIA-RT comes with various AI models for face detection, vehicle and people detection, crowd estimation, and more.
- **Build video analytics workflow in a no-code way.:** Effortlessly create custom video analytics solutions using Lumeo's innovative drag and drop tools.
- **End to end Computer Vision Development:** Our wiki help you start with the computer vision software and development tools during each step of your application.
- **obtain data for training**
- **label data for dataset preparation**
- **train AI models**
- **optimize models to run the best on the edge device**
- **deploy the model on the edge**
- **remote management**

## Part List

|                                              |      |
|----------------------------------------------|------|
| reServer Industrial J4012 (System Installed) | x 1  |
| DIN rail bracket                             | x 1  |
| Bracket screw                                | x 10 |
| 16-Pin Terminal block for DIO                | x1   |
| 24V/5A Power adapter (without power cord)    | x 1  |
| 2-Pin Terminal block power connector         | x 1  |

## Compliance & Logistics



|          |            |
|----------|------------|
| HSCODE   | 8471504090 |
| USHSCODE | 8517180050 |
| UPC      |            |
| EUHSCODE | 8471707000 |
| COO      | CHINA      |

### FCC Requirement

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) this device may not cause harmful interference, and
- (2) this device must accept any interference received, including interference that may cause undesired operation.

15.21

This transmitter must not be co-located or operating in conjunction with any other antenna or transmitter.

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference in a residential installation.

This equipment generates, uses, and can radiate radio frequency energy.

If not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

## Resource Links

- <https://www.seeedstudio.com/reServer-industrial-J4012-p-5747.html>
- [Datasheet](#)
- [Schematic](#)
- [3D File](#)
- [Assembly Guide](#)
- [Seeed Nvidia Jetson Product Catalog](#)
- [Nvidia Jetson Comparison](#)
- [Nvidia Jetson Product Comparison](#)
- [Seeed Nvidia Jetson Success Cases](#)
- [Seeed Jetson One Pager](#)
- [Flash BSP for Jetson](#)

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